Frequency-Domain Heterogeneous Rank-Entropy Bipolarization Network for Remote Sensing Change Detection

Hang Sun, Runzhou Li, Chenyang Wang, Lefei Zhang, *Senior Member, IEEE*, Dong Ren, and Wenbin Wang

***Abstract*—Recently, deep neural networks have been extensively investigated for remote sensing image change detection and have achieved remarkable performance improvements. However, most change detection methods do not effectively disentangle content and style features, making them vulnerable to domain shifts caused by illumination and seasonal variations, thereby producing pseudo-changes. Moreover, existing methods tend to overfit background cues in uncertain regions, causing the decision boundary to rely excessively on background textures and introducing background-induced spurious correlations. To address these issues, we propose a frequency-domain heterogeneous rank-entropy bipolarization network (FHRB-Net) for change detection. Specifically, we propose a frequency-domain heterogeneous-modulation fusion module (FHFM) that enhances change information and suppresses style interference through heterogeneous frequency-aware modulation, and employs a cosine-gated fusion mechanism to enhance bi-temporal feature consistency and alleviate domain shifts. Furthermore, a rank-entropy bipolarization loss (REB-Loss) is designed to quantify uncertainty from the model's predicted probability map using a rank-based cumulative distribution function and Riemann-sum approximation and to reduce uncertainty by polarizing the probability distribution toward bipolar extremes to suppress background-induced spurious correlations. Experiments on several challenging remote sensing change detection (RSCD) datasets demonstrate that the proposed FHRB-Net yields the best overall performance. The code is publicly available at: https://github.com/lrz2025/FHRB-Net**

***Index Terms*—Causal intervention, change detection, Fourier transform, remote sensing, uncertainty.**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to comparatively analyze multitemporal imagery acquired at different times over the same geographic area, thus identifying regions where surface features have changed [1], [2]. As an important topic in remote sensing, RSCD is essential in practical tasks such as urban planning, deforestation monitoring, disaster assessment, and environmental surveillance, and has therefore attracted widespread attention [3], [4], [5], [6]. However, with increasingly diverse data sources and application scenarios, accurately detecting changed areas remains a challenging problem.

Early change detection methods [7], [8] mainly relied on handcrafted feature representations. Due to limited robustness and insufficient semantic capacity, such features often fail in complex real-world scenes. With the advent of deep learning, weight-sharing Siamese CNNs have been widely adopted to extract bi-temporal features and quantify temporal differences in feature space. According to how change is measured, existing methods fall into two categories: metric-based [9], [10], [11] and classification-based [12], [13], [14]. Metric-learning approaches detect changes through a learned distance metric computed over paired images, but they often underexploit semantic cues and contextual information in imagery. In contrast, classification-based methods treat change detection as a classification problem by assigning the extracted temporal features to different classes, enabling more effective use of semantic and spatial context. Despite the promising performance of these methods, there are still two unresolved issues that demand further exploration, as follows.

1) Prevailing Methods Often Fail to Effectively Disentangle Content and Style Representations, Which Degrades Detection Accuracy: Over the past decade, image-to-image translation techniques (e.g., generative adversarial network (GAN)-based methods) have been widely introduced into RSCD to reduce cross-domain discrepancies and improve change detection performance [15], [16], [17], [18]. However, these approaches largely rely on translation quality, making the transformed images prone to artifact distortion [see the red box in Fig. 1(a)]. Recently, several advanced RSCD methods have turned to frequency-domain modeling to alleviate the inevitable style shift caused by differences in imaging conditions across period [19], [20], [21]. Nevertheless, most

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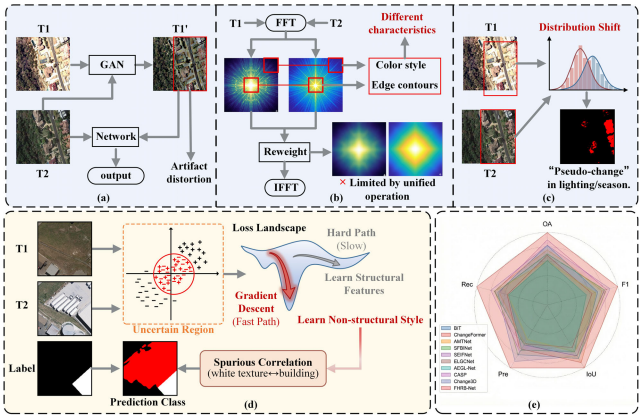


Fig. 1. (a) Style-aligned images obtained via GANs, etc. (b) Filter-based alignment through frequency-domain transformation (FFT). (c) Illustration of pseudo changes. (d) Illustration of spurious correlations. (e) Comparison with state-of-the-art RSCD methods on three popular datasets: LEVIR-CD, SYSU-CD, and WHU-CD.

of these methods simplify frequency-domain processing into a uniform global filtering or style-alignment procedure, and thus fail to fully exploit the distinct and complementary information carried by different frequency components [see Fig. 1(b)]. The aforementioned methods tend to model and correct cross-domain shifts in a unified manner, yet lack effective disentanglement between content and style representations. As a result, in regions with substantial illumination or seasonal variations, these methods are prone to producing pseudo-changes [see Fig. 1(c)]. Therefore, how to learn mutually independent content and style representations while suppressing the interference of imaging-style discrepancies on content cues remains an open problem.

2) How to Mitigate the Model's Overfitting to Background Textures, Achieving More Robust Change Detection? In existing change detection methods, predictions are learned from large-scale data assimilation. There is increasing evidence indicating that, under existing learning paradigms, neural networks inherently tend to prefer highly correlated spurious features during optimization [22], [23], [24], [25]. Although such features are associated with the data labels, they are noncausal. As illustrated in Fig. 1(d), regions with textures similar to the target often exhibit higher uncertainty [26]; these regions are difficult to recognize and tend to incur larger losses. Under gradient-based optimization, learnable parameters preferentially capture features that reduce the loss most rapidly. In fact, background style cues (e.g., brightness or color tone) are often easier to fit than true structural changes, causing the model to prioritize nonstructural cues [27], [28], [29]. As a result, a spurious correlation emerges between imaging style and labels, leading to misclassification when similar styles recur. Hence, effectively preventing overfitting to background appearance and reducing the influence of spurious correlations in uncertain regions remains an open problem.

To address the above-mentioned issues, in this article, we propose a frequency-domain heterogeneous rank-entropy bipolarization network (FHRB-Net) for RSCD. FHRB-Net consists of a frequency-domain heterogeneous-modulation fusion module (FHFM) and a rank-entropy bipolarization loss (REB-Loss). Specifically, FHFM first decouples frequency information via the Fourier transform. For the high-frequency components, we apply differencing to extract change-sensitive signals, and then employ a modulation-and-filtering bottleneck to enhance bi-temporal responses in changed areas while attenuating noise. For the low-frequency components, we use negative cosine similarity to characterize semantic consistency between feature vectors, thereby reducing sensitivity to style variations. The frequency-domain features are then mapped back to the spatial domain, where we further modulate both feature differencing and feature concatenation using cosine similarity, enabling complementary fusion from multiple strategies. Moreover, REB-Loss is motivated by a causal perspective, where imaging styles are treated as confounders. We construct an empirical distribution over model predictions and quantify predictive uncertainty using the cumulative residual entropy, which serves as an observable proxy mediator. By penalizing high-uncertainty regions, REB-Loss intervenes on the backdoor path, alleviates the spurious association between background texture and labels, and encourages the model to learn more intrinsic causal cues. Thus, our FHRB-Net provides a notable improvement in RSCD [see Fig. 1(e)].

The main contributions of this article are as follows.

1) With the carefully designed FHFM, heterogeneous, frequency-aware modulation enables content–style disentanglement, enhancing temporal change cues while suppressing style-induced interference. During feature fusion, multiple constraints on feature differencing are introduced to mitigate the impact of imaging-style variations on texture representations.
2) To alleviate spurious associations induced by style biases, a REB-Loss is proposed to penalize high-uncertainty regions, thus uncovering deeper and more stable causal relationships and eliminating the interference of confounders.
3) We introduce a novel method, FHRB-Net, which integrates the FHFM and REB-Loss. Evaluations conducted on LEVIR-CD, SYSU-CD, and WHU-CD indicate that our approach delivers better performance on remote sensing imagery than multiple recent state-of-the-art methods.

The organization of this article is as follows. Section II reviews related work. Section III details the proposed method, including the FHFM and REB-Loss. Section IV evaluates the method through experiments. Section V discusses the advances of FHRB-Net and future work, as well as the limitations of current evaluation metrics. Finally, Section VI provides the conclusion.

II. Related Work

In Section II, we review the progress of RSCD over the past decades. Section II-A summarizes traditional change detection methods, while Section II-B provides a systematic overview of deep learning-based approaches. Section II-C further discusses recent advances and applications of change detection in the frequency domain.

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A. Traditional Change Detection Methods

Early studies on RSCD primarily relied on hand-crafted features and classical statistical models. Changes between multitemporal images were typically identified by constructing change vectors, spectral indices, or using postclassification comparison to delineate changed and unchanged regions. For example, NDVI [7] and postclassification comparison [8] have been widely adopted in practical applications to localize land-cover changes. However, these approaches are highly dependent on hand-crafted features and the accuracy of the underlying classification; classification errors can accumulate and be amplified during the change detection stage, making it difficult to capture fine-grained or gradual changes.

B. Deep Learning-Based Change Detection Methods

In recent years, deep learning-based RSCD has shown great vitality. Daudt et al. [30] first combined deep learning with RSCD by proposing FC-EF, which employs an end-to-end trained fully convolutional Siamese architecture to directly generate pixel-wise change maps from bi-temporal images. Since then, a large body of work has focused on modeling Siamese encoders. Some works leverage channel and spatial attention mechanisms to improve feature representations [31], [32]. Ying et al. [33] propose AEGL-Net, which extracts bi-temporal multiscale features with a lightweight backbone and applies adaptive multiscale enhancement before fusion.

Nevertheless, simply stacking convolutions or attention blocks still falls short in capturing long-range dependencies and region-level semantic differences in complex scenes. Transformer architectures [34] can efficiently aggregate global context and learn long-range dependencies. Chen et al. [35] propose BIT, which models the spatiotemporal context of bi-temporal images in the token space using a transformer encoder. Bandara and Patel [36] further propose ChangeFormer, a hierarchical transformer encoder that extracts multilevel features at multiple scales and fuses them through a dedicated differencing module. Noman et al. [37] employ a Siamese transformer to extract four-stage multiscale features and fuse them stage by stage at the encoder. Liu et al. [38] integrate spatial attention and transformers to capture position-aware spatial context and global long-range dependencies, respectively.

Meanwhile, diffusion models [39] and foundation models [40] have rapidly advanced in the remote sensing community, providing new paradigms for change detection. Wen et al. [41] present GCD-DDPM, the first purely generative diffusion-based model specifically designed for RSCD. ChangeCLIP [42] is the first to systematically adapt the large-scale vision–language model CLIP to RSCD. Jiang et al. [43], [44] further combine two types of visual foundation models, SAM and CLIP, to extend traditional binary change detection to semantic change detection.

C. Frequency-Based Change Detection Methods

Although existing temporal-domain methods achieve competitive quantitative performance on closed datasets, they often struggle with style discrepancies between bi-temporal images. Motivated by this limitation, recent studies have shifted to frequency-domain representations to suppress style-related perturbations. Liu et al. [21] proposed MDENet, which filters style-related components in the frequency domain and strengthens location-aware attention. Chen et al. [19] introduced FIMP, which suppresses pseudo changes induced by varying imaging conditions through frequency-domain feature interaction coupled with a multiscale perception mechanism. Li et al. [20] proposed FTransDF-Net, leveraging frequency-domain attention to model global dependencies of large-scale targets and thereby mitigate pseudo changes caused by imaging differences. Zang et al. [45] proposed FeaSpect, extending frequency-domain transforms from the pixel space into the feature space.

D. Emerging Change Detection Methods

Beyond the research directions discussed above, several recent studies have proposed new solutions to key challenges in RSCD, further advancing the field. Li et al. [14] were the first to revisit the issue of pseudo changes in building change detection from the perspective of uncertainty modeling. Zhou et al. [46] examined false positives and false negatives in RSCD caused by spectral confusion, shadow occlusion, and related factors from the perspective of deep prompting, and proposed the DepthCD framework. Liu et al. [47] investigated change detection under inconsistent bi-temporal image resolutions from the perspective of arbitrary-resolution change detection. Zhu et al. [48] revisited bi-temporal RSCD and change captioning from the perspective of video modeling, and introduced a unified framework, Change3D. Overall, these recent studies suggest a clear shift in RSCD toward more robust, flexible, and unified frameworks.

III. Methodology

This section is organized as follows. Section III-A details the proposed FHRB-Net, and Section III-B introduces the FHFM. Section III-C presents the REB-Loss, while Section III-D describes the overall loss function.

A. Overview

In this work, we develop an FHRB-Net built upon EDED [49] as the baseline. As presented in Fig. 2, the proposed FHRB-Net is an end-to-end RSCD framework that takes bi-temporal images T_1 and T_2 as input. It comprises two encoders, two decoders, and a dedicated change-detection decoder. Each encoder and decoder is constructed from a stack of convolutional blocks to enable efficient feature extraction. The change-detection branch is designed to emphasize the most informative components across bi-temporal features, thereby facilitating effective fusion of change-related representations and accurately localizing all changed targets.

In FHRB-Net, we introduce an FHFM to alleviate domain shift caused by style discrepancies. To further reduce overfitting to background regions during training, we investigate this issue from a causal perspective and propose a REB-Loss. We then describe FHFM and REB-Loss in detail.

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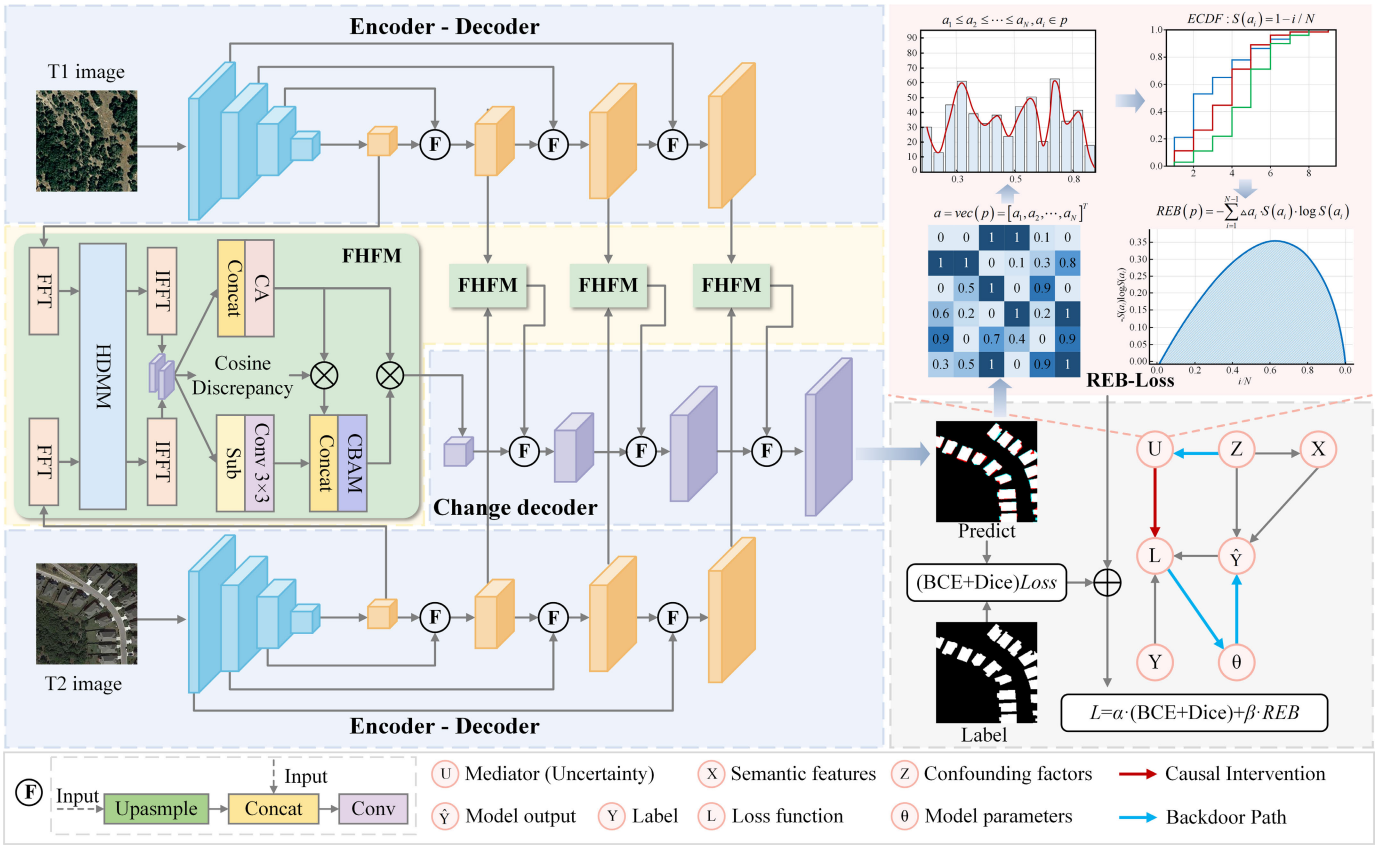


Fig. 2. Overall structure of FHRB-Net. It consists of a dual-branch encoder–decoder and a dedicated change-detection decoder. As the fusion unit between the two branches, FHFM adopts a heterogeneous modulation strategy to accommodate different frequency-domain component properties and integrates them via a cosine-gated fusion mechanism. This design alleviates domain shift caused by style discrepancies. In addition, REB-Loss polarizes the predicted probabilities in highly uncertain regions toward two extremes, thereby suppressing spurious correlations induced by background variations.

B. Frequency-Domain Heterogeneous-Modulation Fusion Module

Existing methods mitigate pseudo changes to some extent by performing global alignment or filtering in the frequency domain [19], [20], [21]. However, they do not exploit the complementary information across different frequency bands to handle style variations between bi-temporal images. In addition, common bi-temporal feature fusion strategies are to use the concatenation or differencing. Such limited fusion not only fails to fully leverage discrepancy cues but also tends to amplify style inconsistencies at the feature level, causing them to be mistaken for genuine semantic changes.

To resolve these issues, we propose an FHFM, which consists of a heterogeneous dual-branch modulation mechanism (HDMM) and a cosine-gated fusion module (CGFM). The overall workflow of FHFM is as follows. Let D_1 and D_2 denote the features obtained from the two decoder branches. We first apply a 2-D Fourier transform to D_1 and D_2 to obtain their frequency-domain representations F_1 and F_2

$$F_i = \mathrm{FFT}(D_i), \quad i \in \{1, 2\}. \tag{1}$$

The frequency-domain features F_1 and F_2 are first heterogeneously modulated by HDMM to obtain the reconstructed frequency representations F'_1 and F'_2. These reconstructed features are then mapped back to the spatial domain via an inverse Fourier transform to produce D'_1 and D'_2

$$D'_i = \mathrm{IFFT}(F'_i), \quad i \in \{1, 2\}. \tag{2}$$

Finally, the modulated features D'_1 and D'_2 are fused by CGFM to obtain the integrated representation D_R.

1) Heterogeneous Dual-Branch Modulation

For frequency-domain features, the low-frequency components are mainly associated with slowly varying intensity information, whereas the high-frequency components emphasize spatial structure [50]. [51] further reveals a strong correlation between frequency components and image style representations. Consequently, an image can be viewed as a superposition of multiple frequency components, each capturing different structures and levels of detail [52]. Leveraging this property, we exploit frequency-domain information to disentangle content and style features. As illustrated in Fig. 3, the HDMM first transforms the input feature maps into the Fourier domain, yielding the frequency-domain representations $\mathbf{F}_1$ and $\mathbf{F}_2$. To explicitly decouple structural and detail information, we construct two fixed complementary masks in the Fourier domain: a low-frequency mask M_L and a high-frequency mask M_H. These masks are generated using a predefined radial frequency threshold centered at the spectrum origin. Specifically, coefficients within the low-frequency region are assigned to M_L, while those outside this region are assigned to M_H. Accordingly, $M_H = 1 - M_L$, and the two masks partition the spectrum into low- and high-frequency components.

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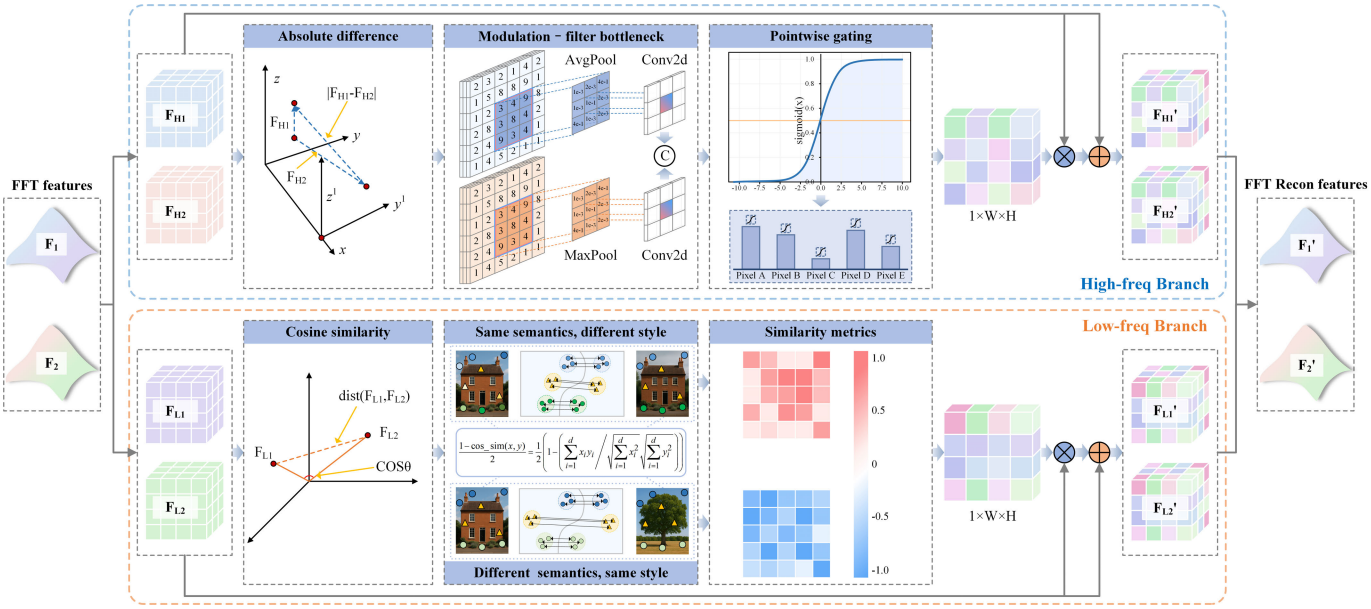


Fig. 3. Overall architecture of HDMM. The high- and low-frequency features obtained via frequency-domain decomposition are fed into the high-frequency and low-frequency branches for modulation, respectively. The high-frequency branch employs a difference-driven modulation–filtering bottleneck to highlight genuine changes while suppressing background texture interference; the low-frequency branch leverages a cosine-similarity constraint and similarity metrics to achieve cross-temporal style alignment, thereby mitigating domain shifts caused by imaging-style variations such as illumination and seasonal changes.

With these masks, $\mathbf{F}_1$ and $\mathbf{F}_2$ are decomposed into the low-frequency components $\mathbf{F}_{L1}$ and $\mathbf{F}_{L2}$, and the high-frequency components $\mathbf{F}_{H1}$ and $\mathbf{F}_{H2}$, respectively, as defined in the following equation:

$$\begin{cases} \mathbf{F}_{L1} &= \mathbf{F}_1 \otimes M_L \\ \mathbf{F}_{L2} &= \mathbf{F}_2 \otimes M_L \\ \mathbf{F}_{H1} &= \mathbf{F}_1 \otimes M_H \\ \mathbf{F}_{H2} &= \mathbf{F}_2 \otimes M_H \end{cases} \tag{3}$$

where $\otimes$ denotes element-wise multiplication. The masks M_L and M_H are real-valued binary masks with the same spatial-frequency resolution as $\mathbf{F}_1$ and $\mathbf{F}_2$, and are broadcast along the channel dimension during implementation. This design allows the masks to be directly applied to the Fourier coefficients in a dimensionally consistent manner.

High-Frequency Branch: High-frequency features mainly correspond to edges, fine structures, and texture variations in the imagery. They delineate object boundaries and structural differences, and thus provide key cues for change detection. However, they are also highly sensitive to noise and background textures, and can easily introduce pseudo-changes under imaging style variations. In the high-frequency branch, we first compute a coarse difference map by performing element-wise subtraction between the high-frequency components $\mathrm{F}_{\mathrm{H}_1}$ and $\mathrm{F}_{\mathrm{H}_2}$ of the bi-temporal features. An absolute operation is then applied to the result to capture the magnitude of discrepancies between the two time points, providing informative cues for identifying changed regions

$$\mathbf{F}_{\text{diff}} = \left|\mathbf{F}_{H1} - \mathbf{F}_{H2}\right| \tag{4}$$

where $|\cdot|$ denotes the element-wise absolute-value operation, and $\mathbf{F}_{\text{diff}}$ denotes the resulting difference feature map used to represent the change regions between the bi-temporal images.

Although the frequency-domain difference feature $\mathbf{F}_{\text{diff}}$ encodes the discrepancies between the bi-temporal features, it also contains substantial redundant background clutter. Therefore, we first apply a 3×3 convolution followed by a normalization operation to preliminarily filter $\mathbf{F}_{\text{diff}}$, and then feed the refined features into the modulation–filtering bottleneck

$$\hat{\mathbf{F}}_{\text{diff}} = \mathrm{PReLU}\big(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{\text{diff}}\right)\right)\big) \tag{5}$$

where $\mathrm{Conv}_{3\times3}(\cdot)$ denotes a 3×3 convolution used for feature extraction and redundancy suppression, $\mathrm{BN}(\cdot)$ denotes batch normalization for stabilizing training and reducing background noise, and $\hat{\mathbf{F}}_{\text{diff}}$ denotes the enhanced difference feature map after convolution, normalization, and activation for subsequent change-region identification.

In the modulation–filtering bottleneck [53], the difference features are first subjected to channel-wise max pooling and average pooling. These pooled descriptors are then passed through a 7×7 convolution followed by a sigmoid function to generate a frequency-domain attention map

$$\mathbf{M}_{\text{diff}} = \sigma\big(\mathrm{Conv}_{7\times7}\left(\left[\,\mathrm{GAP}\left(\hat{\mathbf{F}}_{\text{diff}}\right);\,\mathrm{GMP}\left(\hat{\mathbf{F}}_{\text{diff}}\right)\,\right]\right)\big) \tag{6}$$

where $\mathbf{M}_{\text{diff}}$ denotes the final frequency-domain attention map used to reweight and enhance the difference features.

The preliminarily extracted difference features enhance the responses around sharp boundary transitions, guiding GMP to focus more on edges and locally salient discrepancy regions. In contrast, the global weights produced by GAP help avoid overemphasis on isolated noise and suppress interference from irrelevant areas. Consequently, GMP and GAP complement each other from local and global perspectives, respectively, to derive change-aware weights and enhance the model's sensitivity to complex changes. Finally, the resulting attention

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map is used as a weighting mask and multiplied with the bi-temporal high-frequency features to modulate and filter the high-frequency responses

$$\begin{cases} \mathrm{F}'_{H1} & = \mathbf{F}_{H1} \otimes \mathbf{M}_{\text{diff}} + \mathbf{F}_{H1} \\ \mathrm{F}'_{H2} & = \mathbf{F}_{H2} \otimes \mathbf{M}_{\text{diff}} + \mathbf{F}_{H2} \end{cases}. \tag{7}$$

Low-Frequency Branch: Low-frequency features primarily characterize large-scale, slowly varying components, such as overall brightness and contrast. For the same object in two images, its color may change due to illumination or seasonal differences. If we directly compute the difference on these low-frequency components, the resulting values can be large, even though such changes do not correspond to real semantic variations.

Therefore, we achieve cross-temporal style alignment by computing the cosine similarity between the low-frequency components. Specifically, we first perform L2 normalization along the channel dimension so that all vectors lie on the unit hypersphere, retaining only the directional information while discarding the magnitude

$$\begin{cases} \hat{\mathbf{F}}_{L1} & = \frac{\mathbf{F}_{L1}}{\|\mathbf{F}_{L1}\|_2 + \varepsilon} \\ [3pt]\hat{\mathbf{F}}_{L2} & = \frac{\mathbf{F}_{L2}}{\|\mathbf{F}_{L2}\|_2 + \varepsilon} \end{cases} \tag{8}$$

where $\|\cdot\|_2$ denotes the L2 norm along the channel dimension, ϵ is a small positive constant to ensure numerical stability, and $\hat{\mathbf{F}}_{L1}$ and $\hat{\mathbf{F}}_{L2}$ denote the normalized features. The cosine similarity is then computed as follows:

$$S_L = \frac{\mathbf{F}_{L1}^{\top}\mathbf{F}_{L2}}{(\|\mathbf{F}_{L1}\|_2 + \varepsilon)\,(\|\mathbf{F}_{L2}\|_2 + \varepsilon)} \tag{9}$$

where S_L denotes the cosine similarity between the low-frequency components, which measures the consistency of the bi-temporal low-frequency features.

Cosine similarity measures the directional agreement between two feature vectors in a high-dimensional space. Its values lie in the range $(-1, 1)$ [54]. To make it more suitable as an adaptive weighting factor, we further transform the cosine similarity according to (10). When global low-frequency attributes such as brightness or contrast shift while the underlying structure remains unchanged, the feature magnitudes vary, but their directions in feature space remain nearly unchanged, and the proposed weighting function assigns relatively low weights to these style-related changes. In contrast, when semantic structures are altered, the directions of the feature vectors change noticeably, leading to higher weights that emphasize true semantic variations

$$S_L^{\text{neg}} = \frac{1}{2}\left(1 - \frac{\mathbf{F}_{L1}^{\top}\mathbf{F}_{L2}}{(\|\mathbf{F}_{L1}\|_2 + \varepsilon)\,(\|\mathbf{F}_{L2}\|_2 + \varepsilon)}\right) \tag{10}$$

where S_L^{neg} converts it into a normalized dissimilarity score.

The computed values are then used as weights and multiplied by the bi-temporal low-frequency features to obtain the weighted low-frequency representations

$$\begin{cases} \mathrm{F}'_{L1} & = \mathbf{F}_{L1} \otimes S_L^{\text{neg}} + \mathbf{F}_{L1} \\ \mathrm{F}'_{L2} & = \mathbf{F}_{L2} \otimes S_L^{\text{neg}} + \mathbf{F}_{L2}. \end{cases} \tag{11}$$

This weighting scheme encourages the model to focus more on semantic structural changes rather than global style variations in the low-frequency components, thereby reducing its sensitivity to imaging style.

Algorithm 1 CGFM PseudoCode, PyTorch-Like

Require: D'_1, D'_2.
Ensure: D_R.

```
D_c = Cat(D'_1, D'_2, dim=1).
c1 = Sigmoid(AdaptiveMaxPool2d(D_c).
c2 = Sigmoid(AdaptiveAvgPool2d(D_c).
D_ca = c1 * D_c+ c2 * D_c.
Dsub = Abs(D'_1 − D'_2).
Dsub = Conv(Dsub).
cos_sim = CosineSimilarity(D'_1, D'_2).
m = Unsqueeze(1− cos_sim, dim=1).
D_m=D_ca * m.
D_R = CBAM(Cat(D_m, Dsub, dim=1)) * D_c.
D_R = Conv(D_R).
return D_R
```

2) Cosine-Gated Fusion Module

Existing feature fusion strategies for bi-temporal change detection typically rely on feature concatenation or element-wise subtraction. However, simple concatenation mainly preserves joint semantic context and fails to explicitly emphasize change-sensitive discrepancy information. In contrast, direct subtraction can capture local changes, but it may also amplify appearance variations in unchanged regions, thereby introducing false alarms. To more effectively characterize meaningful discrepancies between bi-temporal features, we introduce a cosine-similarity-based discrepancy measure to complement the subtraction operation. Specifically, cosine similarity reflects the directional consistency between high-dimensional features. We further convert it into a discrepancy cue, termed cosine discrepancy (1 − cosSim), which more explicitly highlights potential change regions with low feature similarity.

Based on these observations, we propose the CGFM, whose pseudocode is given in Algorithm 1. Given the bi-temporal features D'_1 and D'_2, we first concatenate them to obtain a fused semantic representation that preserves the joint contextual information of the two temporal states. Based on this concatenated feature, global average pooling and global max pooling are applied, followed by a sigmoid activation to generate channel-wise attention weights. These weights are used to recalibrate the concatenated feature, enabling the network to adaptively emphasize more informative channel responses before discrepancy-guided modulation. Meanwhile, we compute the absolute feature difference $|D'_1 - D'_2|$ to explicitly capture local change responses, and further refine this difference representation through a convolutional layer.

In parallel, we compute the cosine similarity between D'_1 and D'_2 and transform it into a discrepancy map via cosine discrepancy. This design converts directional inconsistency into an additional change-sensitive cue, thereby complementing the magnitude-based difference branch. The resulting cosine-derived discrepancy map is then used to gate the attention-enhanced concatenated feature, producing a discrepancy-aware semantic representation. Subsequently,

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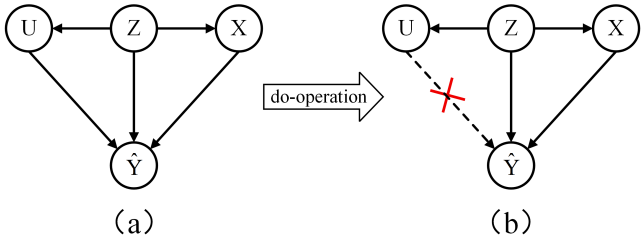


Fig. 4. Proposed structural causal model: (a) the original data-level causal graph, and (b) the intervened graph after applying the do-operation to block the backdoor path and reduce the confounding effect.

the gated feature is concatenated with the refined difference feature and further processed by CBAM, enabling the network to adaptively integrate heterogeneous discrepancy information from different branches. Finally, the refined fusion feature is multiplied element-wise with the concatenated bi-temporal feature, followed by a 1×1 convolution to reduce the channel dimension and produce the final fused representation.

C. Rank-Entropy Bipolarization Loss

In existing deep learning methods, the prediction results are obtained through extensive data fusion. However, such learning paradigms are prone to capturing spurious correlations in the data, rather than uncovering the underlying causal relationships [55]. In change detection, the model tends to overfit the image background in hard-to-discriminate regions, which leads to false associations between background textures and the model D_Rs. Therefore, we revisit the model from a causal perspective to identify the hidden confounders and backdoor paths in change detection, and then perform causal interventions to eliminate the influence of these confounding factors.

1) Structural Causal Model

As shown in Fig. 4(a), we adopt a data-level causal graph to characterize the relationships among the semantic features X, model prediction $\hat{Y}$, and confounder Z, which is induced by imaging style, scene type, and other contextual factors. We further introduce a mediator variable U to capture the uncertainty-related effect of the current sample. However, tend to directly fit the likelihood $P(\hat{Y} \mid X)$ according to Bayes' rule during training. The confounder Z introduces observational bias through $P(Z \mid X)$

$$P\left(\hat{Y} \mid X\right) = \sum_{z \in \mathcal{Z}} P\left(\hat{Y} \mid X, z, M\right) P\left(M \mid X, z\right) P\left(z \mid X\right). \quad (12)$$

It is worth noting that (12) still holds even if we do not explicitly formulate a structural causal model (SCM) for the change detection task, because it is derived from the law of total probability.

The basic principles of the SCM are summarized as follows.

$Z \rightarrow X$: Confounding factors such as imaging style and sensor type determine the appearance distribution of the image, and thus affect the semantic features X extracted by the network.

$Z \rightarrow U$: The same confounders also influence the model's confidence and uncertainty. For example, under rare imaging conditions, the model is more likely to produce unstable predictions. We introduce U to explicitly characterize this uncertainty induced by Z.

$X \rightarrow \hat{Y}$: This is the main causal pathway in conventional prediction models, where the semantic features X determine the final prediction $\hat{Y}$.

$Z \rightarrow U \rightarrow \hat{Y}$: The confounders Z increase the uncertainty U in pseudo-change regions. During backpropagation, these highly uncertain samples dominate the parameter updates, pushing the model to produce spurious predictions $\hat{Y}$ under specific style conditions.

Existing methods, including our FHFM, mainly focus on intervening on the path $Z \rightarrow X$ to reduce the influence of style-related confounders on X, but they often overlook the backdoor path $X \leftarrow Z \rightarrow U \rightarrow \hat{Y}$, which still contains substantial style- and scene-driven spurious correlations [56]. To eliminate this confounding effect, we propose a causal intervention strategy [57] that applies a do-operation on the edge $U \rightarrow \hat{Y}$, as illustrated in Fig. 4(b), thereby blocking the backdoor path $X \leftarrow Z \rightarrow U \rightarrow \hat{Y}$.

2) Causal Intervention Reasoning

Causal Intervention via Training-Oriented Interpretation: To further explain how confounding effects propagate during optimization, we present a training-oriented conceptual interpretation graph in Fig. 5. Unlike the data-level causal graph in Fig. 4(a), this graph is not intended to represent a formal SCM of the data-generating process. Instead, it serves as an explanatory illustration of how background-related bias influences the optimization objective and parameter updates during training.

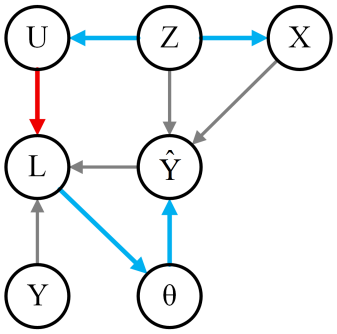


Fig. 5. Conceptual illustration of the intervention mechanism during training. Blue arrows indicate the propagation paths of background-induced bias during training, while red arrows indicate the intervention path introduced by the proposed loss.

Within this interpretive framework, the confounder Z induces higher uncertainty U in pseudo-change regions. As the uncertainty increases, the corresponding cross-entropy loss L and gradients also become larger. During optimization, in order to reduce the loss as quickly as possible, the model preferentially updates the parameters associated with these samples along the path $U \rightarrow L \rightarrow \theta \rightarrow \hat{Y}$. Since nonstructural cues such as imaging style, brightness, or fixed background textures are easier to fit than true structural changes, the model tends to memorize imaging style as if it were genuine semantic change, thereby reinforcing spurious correlations between imaging style and the label Y.

To mitigate this effect, we explicitly introduce an intervention mechanism that targets uncertainty during training.

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Specifically, the proposed REB-Loss suppresses the contribution of highly uncertain regions to the optimization objective, thereby reducing the influence of background-induced bias on parameter updates and improving the model's ability to focus on genuine semantic changes. From this perspective, Fig. 5 illustrates the bias propagation path during training and the intervention path induced by the proposed loss, rather than a strict SCM over data-generating variables. Based on Fig. 5, the following expression can be derived from Bayes' rule:

$$P\left(\hat{Y} \mid do\left(X\right)\right) = \sum_{z \in \mathcal{Z}} P\left(\hat{Y} \mid X, z, U\right) P\left(U \mid X, z\right) P\left(z\right). \quad (13)$$

According to the definition of the do-operation and the refined causal model, we obtain (13). Compared with (12), (13) is no longer influenced by X. Consequently, this intervention explicitly forces the prediction of Y to incorporate each value of Z in a fair manner.

Since $P(Z)$ and $P(U \mid X, Z)$ cannot be explicitly modeled within the algorithm, we instead quantify and intervene on the mediator U. Specifically, REB-Loss is designed to prevent the high uncertainty induced by Z from dominating parameter updates, thereby approximately blocking the backdoor path during optimization.

Rank-Entropy Bipolarization Loss: The computation of REB-Loss proceeds in the following steps. We first flatten the predicted probability map of the model into a 1-D vector

$$\mathbf{a} = \mathrm{vec}\left(\mathbf{p}\right) = \left[a_1, a_2, \ldots, a_N\right]^\top \quad (14)$$

where $a_1 \leq a_2 \leq \cdots \leq a_N$ and $a_i \in \mathbf{p}$, and the sorted vector $\mathbf{a}$ facilitates characterizing the overall distribution of the predicted probabilities.

For each a_i, we compute the empirical cumulative distribution function (ECDF) [58]

$$S\left(a_i\right) = 1 - \frac{i}{N} \quad (15)$$

where $S(a_i)$ represents the proportion of pixels whose predicted probability is not smaller than a_i.

We then define the interval between adjacent prediction values

$$\Delta a_i = a_{i+1} - a_i, \quad i = 1, \ldots, N-1. \quad (16)$$

The term Δa_i is regarded as the "length weight" of the interval and, together with $S(a_i)$, characterizes the overall shape of the probability distribution. Finally, we approximate the integral by a Riemann sum and use it as the final loss function

$$\mathrm{REB}\left(\mathbf{p}\right) = -\sum_{i=1}^{N-1} \Delta a_i \, S\left(a_i\right) \, \ln S\left(a_i\right). \quad (17)$$

This loss can be interpreted as a weighted entropy term over the empirical distribution $S(\cdot)$. When a large number of pixels have predicted probabilities concentrated in the middle range (around 0.5), the model uncertainty U becomes high, the distribution of $S(a_i)$ becomes flatter, and the corresponding entropy term $-S(a_i)\ln S(a_i)$ in the middle region contributes more prominently.

We combine REB with the main loss as an auxiliary term. Taking binary cross-entropy (BCE) as an example, the overall loss is defined as

$$L_{\text{example}} = \alpha \cdot \mathrm{BCE} + \beta \cdot \mathrm{REB}. \quad (18)$$

During backpropagation, the contribution of REB to the gradient with respect to an individual logit z can be approximated by a first-order term

$$\frac{\partial L_{\text{example}}}{\partial z} \simeq \alpha\left(p - y\right) + \beta \frac{\partial \, \mathrm{REB}}{\partial z} \quad (19)$$

where $p = \sigma(z)$ denotes the predicted probability. In this formulation, the standard BCE term is responsible for fitting the labels, while the additional term $\beta\,\partial\mathrm{REB}/\partial z$ specifically penalizes highly uncertain predictions: when $y = 0$, REB encourages p to shrink toward 0; when $y = 1$, it pushes p toward 1. In pseudo-change regions, REB introduces extra gradients that force the model to reduce its reliance on nonstructural cues, such as imaging style and similar scene types, and instead focus on genuine semantic features.

From a causal perspective, by integrating REB into the total loss and imposing additional constraints on high-uncertainty samples at the gradient level, we effectively intervene on the uncertainty mediator U during optimization. This intervention suppresses the influence of the confounder Z on the prediction through the backdoor path $X \leftarrow Z \rightarrow U \rightarrow L \rightarrow \theta \rightarrow \hat{Y}$, thereby guiding the model back to the desired causal relationship and alleviating dataset bias and spurious correlations.

D. Total Loss Function

The main supervision of the model is jointly provided by three branches. Specifically, l_{seg1} and l_{seg2} denote the losses of the upper and lower segmentation branches, respectively, while l_{change} represents the loss of the change-detection branch. For each branch $b \in \{\text{change, seg1, seg2}\}$, we define the branch loss as the sum of a BCE term and a Dice term

$$l_b = \mathcal{L}_{\mathrm{BCE}}\left(\mathbf{P}^{(b)}, \mathbf{Y}^{(b)}\right) + \mathcal{L}_{\mathrm{Dice}}\left(\mathbf{P}^{(b)}, \mathbf{Y}^{(b)}\right) \quad (20)$$

where $\mathbf{P}^{(b)} \in [0,1]^N$ and $\mathbf{Y}^{(b)} \in \{0,1\}^N$ are the predicted probability map and the ground-truth label map, respectively, and N is the number of pixels/voxels.

The BCE loss is defined as

$$\mathcal{L}_{\mathrm{BCE}}\left(\mathbf{P}, \mathbf{Y}\right) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1 - y_i\right)\log\left(1 - p_i\right)\right] \quad (21)$$

and the Dice loss is defined as

$$\mathcal{L}_{\mathrm{Dice}}\left(\mathbf{P}, \mathbf{Y}\right) = 1 - \frac{2\sum_{i=1}^{N} p_i y_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} y_i + \epsilon} \quad (22)$$

where ϵ is a small constant to ensure numerical stability. The main loss is given by

$$L = l_{\text{change}} + 0.2\, l_{\text{seg1}} + 0.2\, l_{\text{seg2}}. \quad (23)$$

Therefore, the final loss function is defined as a weighted sum of the main loss and the REB term

$$L_{\text{total}} = \alpha \cdot L + \beta \cdot \mathrm{REB} \quad (24)$$

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where the scalars α and β control the relative contribution of the two components.

IV. Experiments

In this section, we evaluate the effectiveness of the proposed FHRB-Net on three RSCD datasets: LEVIR-CD, WHU-CD, and SYSU-CD. The datasets and implementation details are introduced in Sections IV-A and IV-B, respectively. Subsequently, we describe the state-of-the-art methods in Section IV-C and provide a performance comparison with these methods in Section IV-D. To quantify the contribution of every component in FHRB-Net, a comprehensive ablation analysis is provided in Section IV-E.

A. Datasets

1) *LEVIR-CD [59]:* This publicly available, large-scale benchmark is designed for building change detection, providing 445, 64, and 128 image pairs for training, validation, and testing, respectively. Each pair consists of two 1024×1024 bi-temporal images. To ensure a fair comparison with previous methods while keeping GPU memory usage manageable, each large image is divided into 16 nonoverlapping patches of size 256×256. In total, we obtain 7120, 1024, and 2048 patch pairs for the training, validation, and test sets, respectively.

2) *WHU-CD [60]:* This dataset is constructed from a pair of large bi-temporal aerial images over Christchurch, New Zealand, with a spatial resolution of approximately 0.2 m. The original imagery covers the earthquake-affected area and the subsequent reconstruction, mainly capturing structural changes such as building construction and demolition. We crop the large images into patches of 256×256 pixels, resulting in 7634 bi-temporal pairs with binary building-change annotations, of which 6096, 762, and 762 pairs are assigned to training, validation, and testing in that order.

3) *SYSU-CD [61]:* This dataset consists of 20 000 pairs of high-resolution aerial images acquired over Hong Kong between 2007 and 2014, with each image of size 256×256 pixels. The dataset encompasses diverse urban change patterns, including new urban constructions, suburban sprawl, preconstruction groundwork, vegetation dynamics, road extensions, and marine engineering. It is therefore highly representative of typical urban change scenarios. Following the official split, the dataset is divided into 12 000, 4000, and 4000 image pairs for training, validation, and testing.

B. Experiment Implementation Details

We implement FHRB-Net in PyTorch. All experiments, including training and inference, are performed on a single NVIDIA RTX 3090 GPU equipped with 24 GB of memory. The model is trained for 200 epochs using a batch size of 8. AdamW is employed as the optimizer with a random seed of 42, an initial learning rate of 2×10^{-4}, and a weight decay of $1S \times 10^{-3}$.

To evaluate the proposed approach, we use five commonly adopted metrics: precision, recall, $F1$-score, overall accuracy (OA), and intersection over union (IoU). The definitions of these metrics are given as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{25}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{26}$$

$$\text{F1} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{27}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{28}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{29}$$

where TP, TN, FP, and FN denote the numbers of correctly detected change pixels, correctly detected unchanged pixels, unchanged pixels incorrectly detected as change, and change pixels incorrectly detected as unchanged, respectively.

C. Comparison With State-of-the-Art Methods

To validate the effectiveness of our method, we compare it with nine state-of-the-art approaches.

1) BiT [35] combines CNN features with transformer attention to enhance spatiotemporal semantics, improving high-level representation utilization for change detection.
2) ChangeFormer [36] adopts a hierarchical Siamese transformer to capture multiscale long-range dependencies and generate pixel-wise change masks.
3) AMT-Net [38] integrates multiscale convolutions and transformers, and reduces cross-temporal discrepancies via partial feature exchange between Siamese branches.
4) SFBI-Net [13] performs same-level global–local bi-temporal interaction and explicitly models spatiotemporal relations, strengthening small-change representation with diversity loss.
5) SEIFNet [62] enhances spatiotemporal differences with interlayer fusion, and improves boundary integrity and intraregion consistency using adaptive context fusion.
6) ELGC-Net [37] aggregates local–global context by combining lightweight convolutions with global attention, enhancing semantic consistency and multiscale contextual dependency.
7) AEGL-Net [33] employs adaptive multiscale enhancement and global–local fusion modules to improve multiscale representation learning and cross-temporal consistency awareness.
8) CASP [3] uses structure-aware fusion with a two-stage alignment strategy to mitigate slight registration errors for more reliable building change representation.
9) Change3D [48] treats bi-temporal images as a short video for unified change detection and description via lightweight video encoding with learnable perceptual frames.

D. Performance Comparison

1) *Analysis of Results on the LEVIR-CD Dataset:* Table I reports the quantitative comparison results on the LEVIR-CD

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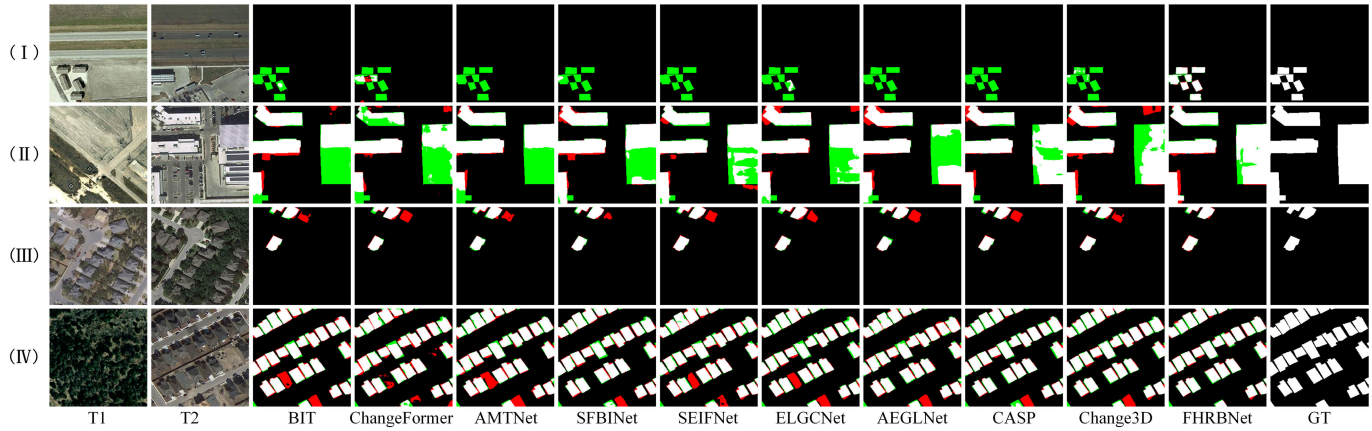


Fig. 6. Qualitative comparisons between the proposed approach and other methods on the LEVIR-CD dataset are presented. In the predicted maps, white denotes true positives, black denotes true negatives, red indicates false positives, and green corresponds to false negatives.

TABLE I

ACCURACY COMPARISON ON THE LEVIR-CD DATASET. THE TOP-PERFORMING RESULTS ARE MARKED IN RED, WHILE THE SECOND-BEST ARE SHOWN IN BLUE. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Methods	OA	F1	IoU	Pre	Rec
BIT	98.98	89.94	81.72	90.32	89.55
ChangeFormer	98.90	89.00	80.18	90.87	87.20
AMTNet	99.04	90.55	82.73	91.26	89.84
SFBINet	99.04	90.48	82.62	92.08	88.94
SEIFNet	98.96	89.58	81.13	92.19	87.12
ELGCNet	99.01	90.18	82.12	91.77	88.65
AEGL-Net	99.12	91.35	84.07	91.92	90.79
CASP	99.12	91.27	83.94	92.51	90.06
Change3D	99.16	91.73	84.72	92.69	90.78
FHRB-Net	99.24	92.45	85.96	93.31	91.60

TABLE II

ACCURACY COMPARISON ON THE SYSU-CD DATASET. THE TOP-PERFORMING RESULTS ARE MARKED IN RED, WHILE THE SECOND-BEST ARE SHOWN IN BLUE. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Methods	OA	F1	IoU	Pre	Rec
BIT	90.10	78.48	64.59	80.51	76.55
ChangeFormer	89.03	76.19	61.54	78.10	74.37
AMTNet	89.96	78.01	63.95	80.67	75.52
SFBINet	90.01	77.51	63.28	82.68	72.95
SEIFNet	91.19	80.72	67.67	83.42	78.19
ELGCNet	90.76	80.21	66.96	81.03	79.40
AEGL-Net	91.78	81.94	69.40	85.00	79.09
CASP	91.59	81.91	69.36	83.11	80.73
Change3D	91.75	82.14	69.70	83.92	80.44
FHRB-Net	92.59	83.84	72.18	86.37	81.45

dataset. As can be seen, the proposed FHRB-Net achieves the best performance on all evaluation metrics: the $F1$, recall, IoU, and precision reach 92.45%, 91.60%, 85.96%, and 93.31%, respectively. The $F1$ and IoU scores are 0.72 and 1.24 percentage points higher than those of the second-best method, Change3D.

Fig. 6 presents representative visual results of different methods on LEVIR-CD. The buildings in this dataset are mostly small- to medium-sized, low-rise structures with regular footprints, dense distribution, and clear boundaries. As shown in Fig. 6(III) and (IV), existing methods generally perform reliably on such "regular" buildings. However, when the scene contains building types that deviate significantly from typical samples, the performance degrades notably. For example, in Fig. 6(I), some buildings exhibit textures that are highly similar to the surrounding pavement, and in Fig. 6(II), parts of the rooftops contain dark regions with colors close to cast shadows. In these cases, most methods fail to correctly separate the buildings from the background and tend to classify them as unchanged regions. The fundamental reason is that these methods heavily rely on statistical appearance correlations learned during training and thus tend to assign regions that "look like the background" directly to the background category, without fully exploiting the stable structural information within those regions. In contrast, FHRB-Net can accurately capture the changed areas of such "atypical" buildings, demonstrating a stronger ability to model the essential differences between buildings and background. This improvement is closely related to the constraint imposed by the proposed REB-Loss on the learning process.

In Fig. 6(III), the same building exhibits pronounced appearance variations between the two time points due to changes in illumination and imaging conditions. Most competing methods yield a large number of false positives in this case, whereas FHRB-Net effectively suppresses pseudo changes caused by brightness and color inconsistencies and focuses more on the building contours and structural shapes. In Fig. 6(IV), some background regions have colors and brightness very similar to those of buildings, and conventional models easily misclassify them as building changes. FHRB-Net, however, can correctly identify these regions as background at the semantic level, leading to more stable change detection results overall.

2) *Analysis of Results on the SYSU-CD Dataset:* As shown in Table II, FHRB-Net achieves the best performance across all metrics and consistently outperforms the competing methods. Specifically, the $F1$ score, recall, IoU, and precision reach 83.84%, 81.45%, 72.17%, and 86.37%, respectively. Compared with the second-best method, Change3D, FHRB-Net improves precision, recall, $F1$, and IoU by 2.45%, 1.01%,

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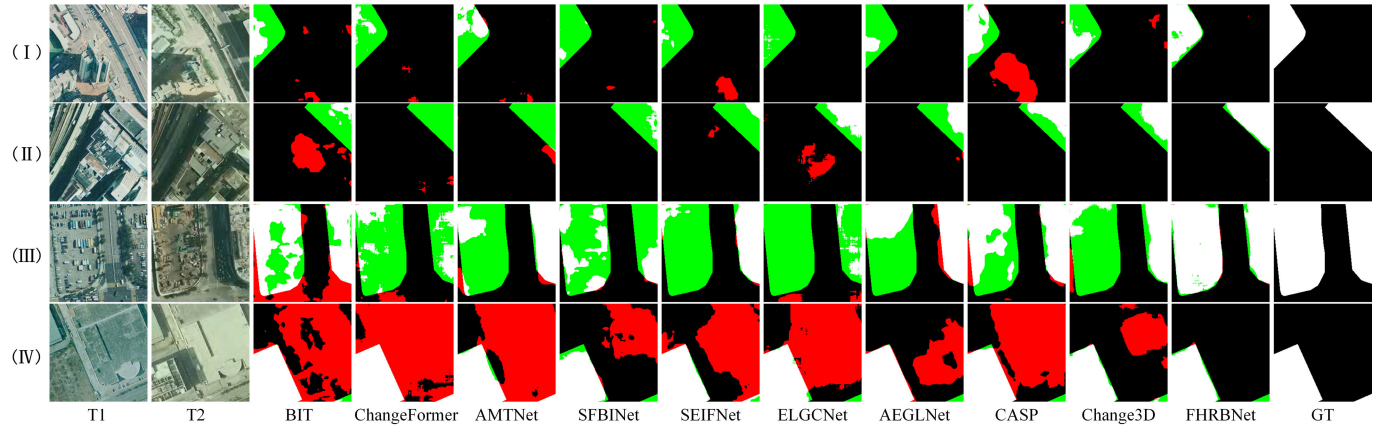


Fig. 7. Qualitative comparisons between the proposed approach and other methods on the SYSU-CD dataset are presented. In the predicted maps, white denotes true positives, black denotes true negatives, red indicates false positives, and green corresponds to false negatives.

2.70%, and 2.48%, respectively, further demonstrating its effectiveness and robustness for change detection.

In the SYSU-CD dataset, the bi-temporal images exhibit pronounced differences in color, brightness, and overall style, and the visual results in Fig. 7 further highlight these challenges. In Fig. 7(I), the target building in the T1 image has textures highly similar to its surrounding background, causing most methods to misclassify it entirely as background and severely miss the true change area. In Fig. 7(II), the background region in the upper right part of the T2 image is extremely similar to buildings in both texture and color, leading many methods to classify this region as buildings and consequently overlook the actual changed area.

In Fig. 7(III), the left region of T2 has high information entropy and complex textures, making its semantic category difficult to determine, while its overall color style is still close to the surrounding background. Conventional methods that depend mainly on shallow statistical features tend to incorrectly assign this region to the background. In all of these complex scenarios, FHRB-Net can still accurately distinguish buildings from background, demonstrating a stronger capability to perceive semantic information. This advantage is closely related to the joint effect of FHFM and REB-Loss: FHFM enhances the differences in feature space that correspond to true structural changes between the two time points, while suppressing the interference caused by style inconsistencies; REB-Loss imposes constraints on highly uncertain regions, thereby reducing erroneous responses in confusing areas. In Fig. 7(IV), the same building appears with markedly different colors across the two time points due to imaging style variations. Simple difference-based fusion strategies typically produce numerous false detections in this region. Benefiting from CGFM, which captures similarity among high-dimensional features and compensates for the limitations of pure difference measures, FHRB-Net is still able to robustly preserve the structural consistency of buildings and accurately localize the true changed regions.

3) *Analysis of Results on the WHU Dataset:* As shown in Table III, FHRB-Net attains the best performance across all evaluation metrics on WHU-CD. Change3D also performs competitively on this dataset, which is closely related to the characteristics of WHU-CD: it mainly focuses on newly constructed and demolished buildings in postearthquake and reconstruction scenarios, where the geometric structures of building edges and roof outlines exhibit pronounced temporal changes. Such strong structural signals are relatively easy for the video encoder to capture.

TABLE III
ACCURACY COMPARISON ON THE WHU-CD DATASET. THE TOP-PERFORMING RESULTS ARE MARKED IN RED, WHILE THE SECOND-BEST ARE SHOWN IN BLUE. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Methods	OA	F1	IoU	Pre	Rec
BIT	98.76	85.74	75.03	80.89	91.20
ChangeFormer	98.64	81.28	69.90	85.37	79.41
AMTNet	99.30	91.14	83.73	91.50	90.78
SFBINet	99.34	91.52	84.37	94.65	88.60
SEIFNet	99.05	88.07	78.68	88.38	87.76
ELGCNet	99.33	91.50	84.33	92.72	90.31
AEGL-Net	99.52	93.81	88.35	96.25	91.50
CASP	99.53	93.32	87.47	95.80	90.96
Change3D	99.56	94.42	89.43	95.97	92.93
FHRB-Net	99.59	94.72	89.96	96.33	93.17

Fig. 8 shows representative visual results of different methods on the WHU test set. Compared with SYSU-CD, WHU-CD has more stable imaging conditions and smaller radiometric differences between the two time points. However, buildings and backgrounds often share similar color and texture styles. For example, in the right-hand region of T1 in Fig. 8(I) and the right-hand region of T2 in Fig. 8(II), the visual characteristics of buildings and ground are highly similar, leading many methods to produce large-scale false positives in these areas. In Fig. 8(I), although ChangeFormer yields relatively few false alarms in nontarget regions, it completely overlooks the large target building block on the right, effectively classifying the entire area as background.

In Fig. 8(III), the changed region is almost indistinguishable in appearance from background shadows and is easily misinterpreted as the same land-cover type. Similarly, in Fig. 8(IV), the changed region on the left and the background on the right exhibit very similar visual styles in both T1 and T2. For such scenes, conventional methods tend to over-reliance on

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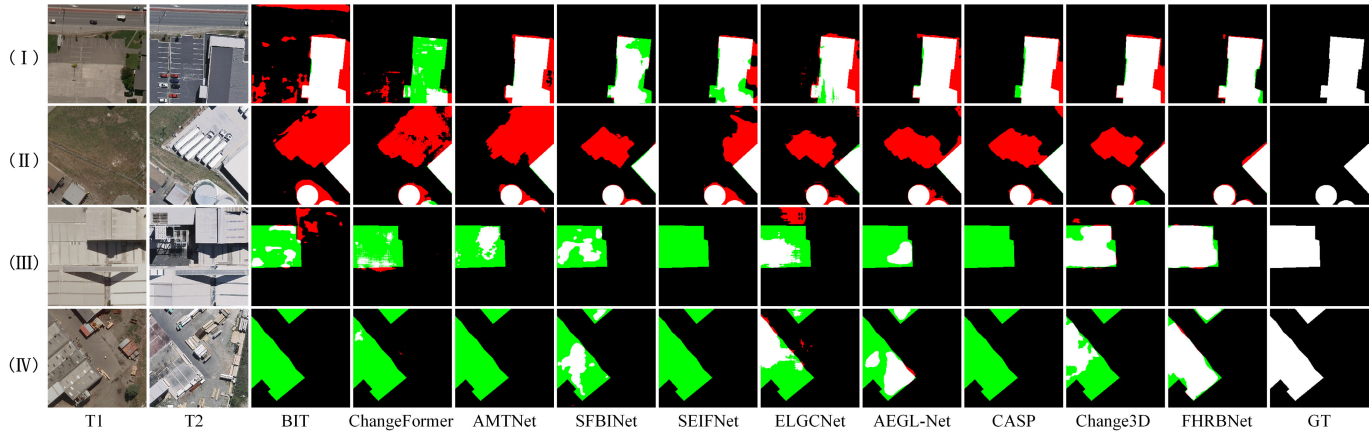


Fig. 8. Qualitative comparisons between the proposed approach and other methods on the WHU-CD dataset are presented. In the predicted maps, white denotes true positives, black denotes true negatives, red indicates false positives, and green corresponds to false negatives.

TABLE IV
ABLATION STUDY RESULTS OF FHFM AND REB-LOSS IN FHRB-NET ON THREE DIFFERENT DATASETS. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Index	Baseline	FHFM	REB-Loss	LEVIR-CD		SYSU-CD		WHU-CD	
				F1 Score	IoU	F1 Score	IoU	F1 Score	IoU
1	√	×	×	91.27	83.94	82.51	70.23	93.71	88.16
2	√	√	×	92.16	85.46	83.51	71.69	94.13	88.91
3	√	×	√	92.08	85.32	83.14	71.14	94.49	89.56
4	√	√	√	**92.45**	**85.96**	**83.84**	**72.18**	**94.72**	**89.97**

color and brightness differences and struggle to extract reliable semantic cues. In contrast, FHRB-Net is able to focus on building contours and stable structural patterns in these challenging cases, effectively suppressing the interference caused by appearance similarity. As a result, it can still accurately segment the true change regions under complex backgrounds, while significantly reducing both false positives and missed detections.

E. Ablation Study

To verify the effectiveness of each component in FHRB-Net, we conduct ablation studies on three datasets under the same experimental settings. Starting from the baseline model (EDED), which adopts ADD as the feature fusion strategy, we progressively introduce FHFM and REB-Loss. Their inclusion is indicated in Table IV by "✓" and "×." Specifically, Index 1 denotes the baseline, Index 2 denotes baseline + FHFM, Index 3 denotes baseline + REB-Loss, and Index 4 denotes the complete FHRB-Net with both FHFM and REB-Loss. Table IV reports the quantitative results in terms of $F1$ score and IoU, with the best performance highlighted in bold.

The quantitative results clearly show the contribution of each component. Compared with Index 1, introducing FHFM alone (Index 2) improves the $F1$ score by 0.89%, 1.00%, and 0.42%, and the IoU by 1.52%, 1.46%, and 0.75% on the three datasets, respectively. Similarly, adding REB-Loss alone (Index 3) also yields consistent improvements: the F1 score increases by 0.81%, 0.63%, and 0.78%, while the IoU increases by 1.38%, 0.91%, and 1.40%, respectively. When both FHFM and REB-Loss are incorporated (Index 4), FHRB-Net achieves the best performance on all three datasets, demonstrating that each component contributes significantly to the detection accuracy.

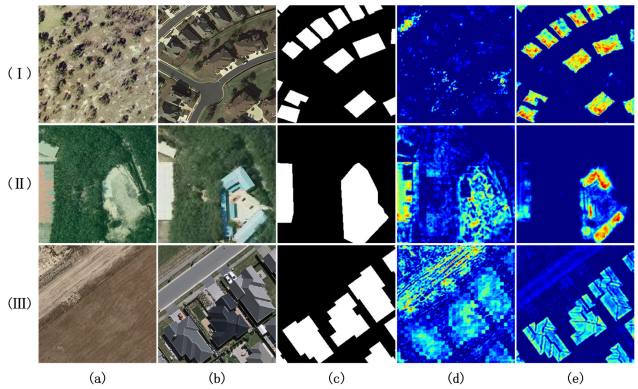


Fig. 9. Partial visualizations of feature maps obtained from the FHFM ablation study on three datasets are shown. (Top to bottom) Three rows correspond to LEVIR-CD, SYSU-CD, and WHU-CD, respectively. (a) T1 image. (b) T2 image. (c) Label. (d) Before FHFM. (e) After FHFM.

Although these results quantitatively confirm the effectiveness of FHFM and REB-Loss, they do not directly illustrate how they improve the final change maps. Therefore, we further perform a qualitative analysis by combining multiple ablation settings with corresponding heatmaps.

1) *Effectiveness of FHFM:* As illustrated in Fig. 9, we use class activation maps (CAMs) to visualize the bi-temporal feature maps before and after modulation by the FHFM. CAMs provide heatmaps that intuitively reflect the correlation between image regions and the change class: warmer colors indicate higher relevance, whereas cooler colors indicate lower relevance.

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TABLE V

ABLATION EXPERIMENTS ON FEATURE FUSION METHODS ON LEVIR-CD. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Fusion Methods	OA	F1	IoU	Pre	Rec
Add	99.20	92.08	85.32	92.93	91.24
Concat	99.21	92.12	85.39	93.19	91.07
A - B	99.20	92.01	85.20	93.22	90.82
|A - B|	99.18	91.83	84.89	92.87	90.81
TFAM	99.21	92.16	85.46	93.06	91.27
TFIM	99.21	92.21	85.55	93.10	91.34
DFC	99.21	92.18	85.49	92.56	**91.80**
FHFM	**99.24**	**92.45**	**85.96**	**93.31**	91.60

In Fig. 9(I), for the LEVIR-CD dataset, the model without FHFM pays relatively little attention to the actual change regions and fails to clearly separate building changes from the background. After introducing FHFM, the responses over the true change regions are significantly strengthened. In Fig. 9(II), for the SYSU-CD dataset, noticeable color inconsistencies in vegetation across the bi-temporal images cause the encoder to incorrectly treat some vegetation as change regions, thereby contaminating the training. After modulation by FHFM, these pseudo-changes are effectively suppressed. In Fig. 9(III), for the WHU-CD dataset, the model without FHFM mistakenly regards roads and some background areas as correlated with the change category. With FHFM, background style is better filtered out, and the model focuses more accurately on the true change regions. These observations further confirm the effectiveness of FHFM in suppressing background style and highlighting real changes.

In addition, we conducted a systematic comparison of several bi-temporal feature fusion strategies, as summarized in Table V. Specifically, "Add" denotes element-wise addition of bi-temporal features, "Concat" represents channel-wise concatenation, while "$A-B$" and "$|A-B|$" correspond to the direct feature difference and its absolute difference, respectively. We further included three representative classical fusion modules, namely, TFAM, TFIM, and DFC, to provide a more comprehensive evaluation of fusion effectiveness.

As shown in Table V, conventional fusion strategies such as Add, Concat, and difference-based fusion ($A - B$ and $|A - B|$) achieve relatively comparable overall performance. However, using only $|A - B|$ leads to noticeable degradation in F1 score and IoU, indicating that relying solely on a single difference representation is insufficient for accurate and fine-grained change recognition. In contrast, the introduced classical fusion modules, including TFAM [49], TFIM [63], and DFC [42], improve the performance to some extent, demonstrating the effectiveness of more elaborate feature interaction mechanisms. Nevertheless, the proposed FHFM achieved the best overall performance among all compared methods, with an $F1$ score of 92.45% and an IoU of 85.96%. These results demonstrate that the integration of heterogeneous modulation and multistrategy fusion effectively overcomes the limitations of single-operation fusion strategies and more fully exploits the complementarity of semantic information from bi-temporal images. Moreover, FHFM suppresses pseudo changes more effectively, thereby significantly enhancing the stability and accuracy of change detection.

TABLE VI

ACCURACY ANALYSIS OF CGFM ON THE LEVIR-CD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Methods	F1	OA	Pre	Rec	IoU
NO CA	92.29	99.22	92.49	**92.10**	85.68
NO Concat+CA	92.03	99.19	92.41	91.66	85.24
NO CD	92.36	99.22	92.68	92.04	85.80
NO Sub	92.28	99.22	93.18	91.4	85.67
CGFM	**92.45**	**99.24**	**93.31**	91.6	**85.96**

To further investigate the effectiveness of FHFM from a structural perspective, we conducted an ablation study on its core component, CGFM, and report the results in Table VI. Specifically, "NO CA" denotes removing the channel attention mechanism, "NO Concat+CA" denotes removing the attention-enhanced concatenation branch, "NO CD" denotes removing the cosine-discrepancy-based modulation, and "NO Sub" denotes removing the difference branch. The complete CGFM achieves the best overall performance, with an $F1$ score of 92.45% and an IoU of 85.96%, while all reduced variants exhibit performance degradation to different degrees. Specifically, the channel attention mechanism and the attention-enhanced concatenation branch help preserve and strengthen the joint semantic context of bi-temporal features, thereby improving the robustness of the fused representation. The cosine-discrepancy-based modeling introduces additional discrepancy cues that effectively complement direct differencing, while the explicit difference branch plays an important role in characterizing local change regions. Overall, these results fully validate the effectiveness of all components in CGFM.

2) *Effectiveness of REB-Loss:* We preconstruct and keep fixed two types of region masks: a core region mask R_c and a style region mask R_s. The core mask R_c is obtained by applying one morphological dilation to the ground-truth change mask $Y \in \{0,1\}^{H\times W}$. It covers the changed area and its boundary neighborhood, enabling a more stable characterization of the underlying causal structure. The style mask R_s captures high-uncertainty regions where the model is prone to misclassification. Specifically, we first train a baseline model to convergence and obtain pixel-wise predicted probabilities p_{ij} on the validation set. Among background pixels with $Y_{ij} = 0$, we select background pixels with high predicted change probabilities using a predefined threshold, and define R_s as the mask over the selected pixels

$$C_s = \{(i,j) \mid p_{ij} \geq \tau_p\}. \tag{30}$$

The threshold is set to $\tau_p = 0.4$. We then apply a morphological closing followed by a dilation to C_s, producing a more connected and robust style region R_s. Finally, we use the same (R_c^m, R_s) as fixed masks in all ablation settings, ensuring that performance differences reflect the true contribution of REB-Loss. To further validate the effectiveness of REB-Loss, we conduct a channel-wise core–spurious analysis. Specifically, we take the last-layer feature map $F \in \mathbb{R}^{C\times H'\times W'}$ and the channel weights of the prediction head $w \in \mathbb{R}^C$. We downsample the predefined R_c and R_s to the feature-map resolution using nearest-neighbor interpolation, yielding R_c' and R_s'. For

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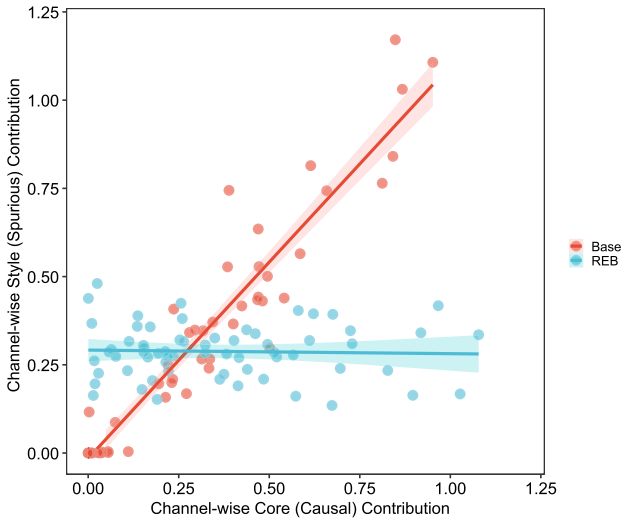


Fig. 10. Scatter plot illustrates the channel-wise core (causal) contribution versus spurious (style) contribution on LEVIR-CD, where each point denotes one feature channel.

each channel k, we compute the mean activation within R'_c and R'_s, denoted as $A_c^{(k)}$ and $A_s^{(k)}$, and weight them by $|w_k|$ to obtain the channel contribution

$$I_c^{(k)} = |w_k| A_c^{(k)}, \qquad I_s^{(k)} = |w_k| A_s^{(k)}. \tag{31}$$

We then plot a scatter diagram using the following equation to compress the dynamic range of channel contributions, preventing a few extremely large channels from dominating the analysis:

$$(x_k, y_k) = \left(\log\left(1 + I_c^{(k)}\right), \log\left(1 + I_s^{(k)}\right)\right). \tag{32}$$

Fig. 10 compares the distributions of channel-wise core and style contributions for the baseline model and the model trained with REB-Loss. Each point corresponds to one channel, where the x-axis and y-axis denote the channel's weighted contribution within the core region R_c and the style region R_s, respectively. For the baseline, points are largely concentrated around the diagonal $y = x$, indicating a similar reliance on core structure and style background, with some channels exhibiting stronger style contributions.

After introducing REB-Loss, the scatter shifts downward, and more channels fall below the diagonal, suggesting that REB-Loss reduces the extent to which gradients are driven by the style region and encourages channels to contribute more within the core region. A small number of channels in the lower left still respond more strongly in R_s than in R_c, but their overall weighted contributions remain small and thus have limited impact on the final prediction. These channels typically capture easily separable background patterns with low discriminative difficulty, and retaining a weak style response helps preserve basic background discrimination. Overall, this analysis provides channel-level evidence that REB-Loss effectively reduces the model's reliance on spurious shortcuts and promotes the dominance of core features.

To further evaluate the effectiveness of REB-Loss, we fixed $\alpha = 1$ in all experiments and used β to control its contribution. The coefficient α was fixed because it serves as the reference weight of the main loss, while the influence of REB-Loss is governed by the relative weighting introduced by β. We then performed a sensitivity analysis with $\beta \in \{0, 0.05, 0.1, 0.25, 0.5, 0.75, 1\}$. As shown in Table VII, the best overall performance is achieved at $\beta = 0.1$, with $F1$, OA, and IoU reaching 92.45, 99.24, and 85.96, respectively. In contrast, performance drops slightly when β is set either too small or too large. A possible reason is that BCE and Dice losses dominate detection accuracy, whereas REB-Loss mainly acts as an auxiliary regularizer for uncertainty reduction. When β becomes too large, the optimization is biased toward REB-Loss, weakening the balance between the primary and auxiliary objectives. This may lead to overly extreme output probabilities, reduced discriminability, and increased overconfidence, which is also supported by the decrease in recall. Accordingly, $\alpha = 1$ and $\beta = 0.1$ are used as the default settings in all experiments.

TABLE VII

ACCURACY COMPARISON ON THE LEVIR-CD DATASET UNDER FIXED $\alpha = 1.00$: SENSITIVITY ANALYSIS RESULTS WITH RESPECT TO β. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

β	F1	OA	Pre	Rec	IoU
0	92.16	99.21	93.06	91.27	85.46
0.05	92.22	99.22	93.51	90.97	85.56
0.10	**92.45**	**99.24**	**93.31**	**91.60**	**85.96**
0.25	92.35	99.23	93.29	91.44	85.79
0.50	92.26	99.22	93.06	91.47	85.63
0.75	92.17	99.22	93.85	90.55	85.48
1.00	92.07	99.21	93.70	90.50	85.31

F. Computational Complexity Analysis

To further assess computational efficiency, Table VIII compares the proposed method with ten competing approaches in terms of parameters, FLOPs, inference time, and FPS. AEGL and CASP adopt lightweight backbones, namely, MobileNetV2 and CASP-Mb0, which reduce model size and accelerate inference, but sacrifice accuracy. Change3D achieves the second-best overall performance, yet its advantage comes at the cost of substantially higher computational complexity, as indicated by its large FLOPs and slower inference speed. In contrast, FHRB-Net delivers the best overall performance with only moderate model complexity, requiring 24.88-M parameters and 20.49 GFLOPs, while achieving an inference time of 9.77 ms and 102.37 FPS. These results demonstrate that FHRB-Net achieves a superior trade-off between accuracy and efficiency compared with existing methods.

V. DISCUSSION

In the above experiments, LEVIR-CD, SYSU-CD, and WHU-CD mainly focus on building-related changes. To further validate the effectiveness of the proposed model in other scenarios and to assess its robustness to variations in imaging styles, we conduct additional experiments on the CDD dataset [64]. CDD contains 11 pairs of bi-temporal remote sensing images with pronounced seasonal differences and covers diverse land-cover types, including farmland, buildings, roads,

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TABLE VIII
COMPARISON OF FLOPS AND PARAMETERS BETWEEN THE PROPOSED METHOD AND TEN COMPARISON METHODS

Methods	Params(M)↓	FLOPs(G)↓	Inf.Time (ms)↓	FPS↑	LEVIR-CD		SYSU-CD		WHU-CD	
					F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
BIT	12.40	10.63	3.96	252.72	89.94	81.72	78.48	64.59	85.74	75.03
ChangeFormer	29.75	21.18	23.60	42.37	89.00	80.18	76.19	61.54	81.28	69.90
AMTNet	24.88	21.56	8.32	120.18	90.55	82.73	78.01	63.95	91.14	83.73
SFBINet	19.08	45.20	19.65	50.89	90.48	82.62	77.51	63.28	91.52	84.37
SEIFNet	28.60	8.37	12.35	81.00	89.58	81.13	80.72	67.67	88.07	78.68
ELGC	10.57	123.59	42.48	23.54	90.18	82.12	80.21	66.96	91.50	84.33
AEGL-Net	**3.19**	**2.35**	21.23	47.09	91.35	84.07	81.94	69.40	93.81	88.35
CASP	4.57	2.63	**1.45**	**690.87**	91.27	83.94	81.91	69.36	93.32	87.47
Change3D	6.43	132.67	8.36	119.59	91.73	84.72	82.14	69.70	94.42	89.43
FHRB-Net	24.88	20.49	9.77	102.37	**92.45**	**85.96**	**83.84**	**72.18**	**94.72**	**89.97**

TABLE IX
ACCURACY COMPARISON ON THE CDD DATASET. THE TOP-PERFORMING RESULTS ARE MARKED IN RED, WHILE THE SECOND-BEST ARE SHOWN IN BLUE. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Methods	OA	F1	IoU	Pre	Rec
BIT	98.86	95.16	90.77	95.91	94.42
ChangeFormer	95.13	77.98	63.90	83.72	72.97
AMTNet	98.83	95.04	90.56	95.06	95.02
SFBINet	99.01	95.78	91.90	96.38	95.18
SEIFNet	98.66	94.26	89.15	95.42	93.14
ELGCNet	99.29	97.01	94.21	96.99	97.04
AEGL-Net	99.21	96.93	94.04	97.36	96.50
CASP	99.20	96.62	93.47	96.69	96.56
Change3D	99.17	96.78	93.77	97.09	96.48
FHRB-Net	99.45	97.87	95.79	98.36	97.38

TABLE X
CROSS-DATASET GENERALIZATION RESULTS OF ALL METHODS TRAINED ON LEVIR-CD AND TESTED ON WHU-CD. ALL METRICS ARE REPORTED AS PERCENTAGES (%)

Methods	OA	F1	IoU	Pre	Rec
BIT	97.10	67.63	51.09	60.68	76.37
ChangeFormer	90.35	33.91	20.42	23.28	62.39
AMTNet	96.61	64.98	48.13	55.11	79.15
SFBINet	97.88	67.93	51.43	84.85	56.64
SEIFNet	97.44	63.53	46.55	73.07	56.19
ELGCNet	97.46	69.44	53.19	66.60	72.53
AEGL-Net	97.81	70.05	53.91	76.72	64.46
CASP	97.53	57.54	40.39	90.66	42.15
Change3D	98.01	74.61	59.50	75.63	73.61
FHRB-Net	98.40	78.19	64.19	85.39	72.11

and vegetation. This dataset, therefore, provides a suitable benchmark for evaluating the impact of seasonal appearance variations.

As presented in Table IX, the proposed method delivers the top results on CDD for all evaluation metrics, with an $F1$-score of 97.87%, a recall of 97.38%, an IoU of 95.79%, and an OA of 98.36%. Compared with the second-best method, Change3D, FHRB-Net improves $F1$, recall, IoU, and OA by 1.09%, 0.90%, 2.02%, and 1.27%, respectively. These results demonstrate that FHRB-Net maintains strong change detection performance and good generalization ability even in complex scenes characterized by significant seasonal variation and a mixture of multiple land-cover categories.

Beyond robustness to seasonal appearance variations, we further evaluate the generalization ability of FHRB-Net through cross-dataset testing. Specifically, the model is trained on LEVIR-CD and directly tested on WHU-CD, where both datasets target building change detection. As shown in Table X, FHRB-Net achieves the best performance on the key metrics, i.e., $F1$-score and IoU. Compared with the second-best method, Change3D, FHRB-Net improves $F1$-score and recall by 3.58% and 4.69%, respectively, demonstrating stable performance under dataset shift. In contrast, other competing methods (e.g., Change3D) exhibit a more notable performance drop under the same setting. Although CASP attains relatively high OA, its recall decreases substantially, indicating reduced sensitivity to changed regions. Overall, these results further confirm the strong cross-domain generalization of FHRB-Net and highlight its potential for large-scale, real-world change detection using optical remote-sensing imagery.

VI. CONCLUSION

In this article, we proposed FHRB-Net for RSCD. First, the FHFM module modulates bi-temporal features in the frequency domain, explicitly enhancing temporal change information while suppressing background interference. Then, in the spatial domain, multiple feature fusion strategies are employed to align feature styles, effectively alleviating domain shifts caused by differences in imaging styles. From a causal perspective, we further analyzed how confounding factors interfere with the model's decision process and designed REB-Loss to intervene in the predicted probability distribution, thereby suppressing background-induced spurious correlations. Extensive experiments on four publicly available change detection datasets show that FHRB-Net consistently outperforms nine state-of-the-art change detection methods. Future work will further explore domain-generalizable change detection methods.

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